

# Richard Liu

437-360-8862 | [richardliu715@gmail.com](mailto:richardliu715@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## EDUCATION

### University of Toronto

September 2021 – May 2026

*Bachelor of Computer Engineering (With Honours); Minor in Artificial Intelligence*

*Toronto, ON*

## SKILLS

**Languages:** C, C++, C#, Python, SQL, Java, JavaScript, HTML, CSS

**Firmware & Embedded Systems:** STM32, ARM Cortex-M3, FreeRTOS, W5500, Modbus RTU, BACnet/IP, Hardware Peripherals (GPIO, UART/USART, SPI, timers, PWM, DMA, interrupts)

**Technologies:** CUDA, Ethernet, TCP/UDP, Docker, Kubernetes, REST APIs, CI/CD

**Developer Tools:** Git, Linux, Keil uVision, J-Link, Wireshark, YABE, Tera Term, VS Code, Bash, Jenkins, Azure

## EXPERIENCE

### Firmware Engineer

May 2026 – August 2026

*Miniveil Systems*

*Toronto, ON*

- Developed bare-metal C firmware for an STM32F103 ARM Cortex-M3 controller, integrating Ethernet, serial communication, timing, and control peripherals.
- Implemented six Modbus RTU function codes over UART with CRC-16 validation, mapping device registers to fan, heat, fault, delay, and status controls.
- Integrated a BACnet/IP server on a memory-constrained STM32 microcontroller, enabling Who-Is/I-Am discovery and ReadProperty communication over W5500 Ethernet and UDP.
- Resolved compiler, linker, protocol, and hardware integration failures by tracing defects across application, transport, driver, and physical layers.
- Debugged end-to-end communication using UART traces, Wireshark, YABE, Tera Term, and J-Link across application, protocol, transport, driver, and hardware layers.

### Software Engineer

May 2024 – August 2025

*Ontario Teachers' Pension Plan*

*Toronto, ON*

- Architected a C# and Python K-1 document processor using Microsoft Document Intelligence, reducing a multi-day manual workflow to under one hour.
- Refactored four applications into one Python microservice, cutting runtime from 20 minutes to 2 minutes while eliminating failure points.
- Built a secure credential-rotation service using Python, REST APIs, CyberArk, RSA-OAEP encryption, and managed secrets for automated production access.
- Delivered software across design, implementation, testing, code review, CI/CD, deployment, documentation, and stakeholder demonstrations in production.

## PROJECTS

### CUDA-Accelerated LiDAR Voxelization | C++, CUDA, Python

August 2026

- Developed CPU and GPU LiDAR voxelization pipelines in C++ and CUDA to convert KITTI point clouds into sparse 3D occupancy grids.
- Implemented CUDA voxelization with spatial filtering and 3D-to-1D indexing.
- Benchmarked kernel and end-to-end performance with CUDA events and C++ timers, and built synchronized LiDAR, voxel, and camera playback in Plotly/Dash.

### V2I Shared Perception System | Python, CARLA, YOLO, Distributed Systems

September 2025 – March 2026

- Designed a distributed Vehicle-to-Infrastructure perception platform that fused sensor data from multiple vehicles.
- Implemented point-cloud alignment and YOLO-based object detection to detect obstacles beyond an individual vehicle's line of sight.
- Achieved approximately 10 Hz map fusion and 200 ms end-to-end latency for real-time ADAS decision support.
- Received the **Certificate of Distinction** and placed **2nd overall among Comp Eng capstone projects**.

### GIS Mapping Engine | C++, Dijkstra's, A\*, Graph Data Structures, Priority Queues

January 2024 – April 2024

- Used C++ to implement pathfinding algorithms including Dijkstra's and A\* within a custom GIS system
- Developed data structures for map representation and used OSM API for data acquisition
- Applied spatial gridding and data grouping techniques to optimize frame rates and improve overall map performance.